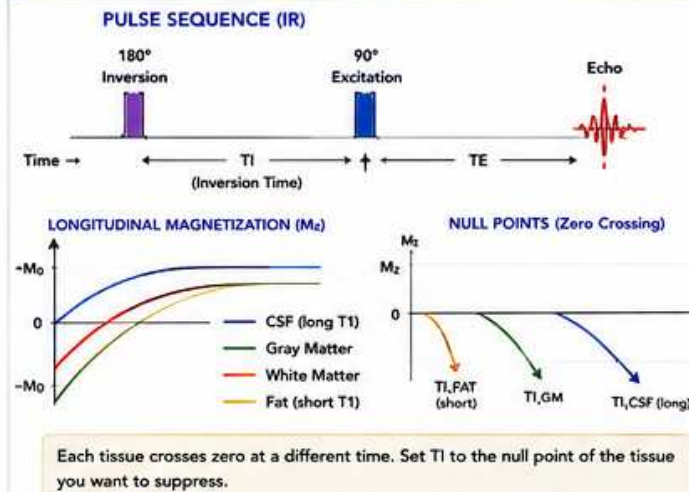


3.3 FLAIR & INVERSION RECOVERY

NULLING UNWANTED TISSUE TO REVEAL PATHOLOGY

FLAIR (Fluid-Attenuated Inversion Recovery) is an Inversion Recovery sequence designed to null (suppress) CSF signal while keeping T2 contrast. Other IR sequences can null different tissues by choosing the right TI.

1 HOW INVERSION RECOVERY WORKS

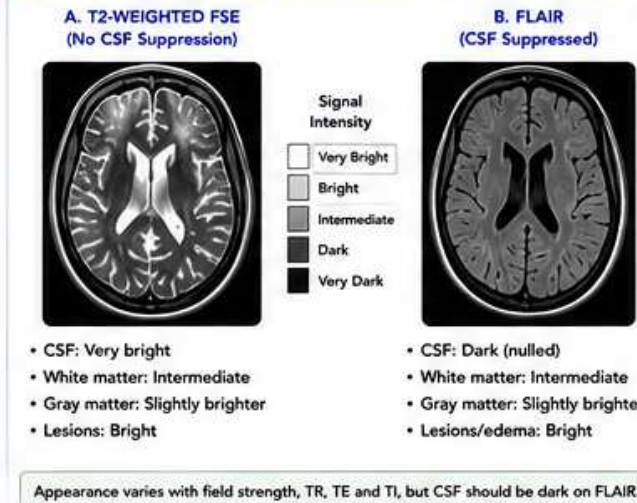


2 WHY THE TECHNOLOGIST USES FLAIR

- ✓ Suppresses CSF (very long T1) → lesions near CSF stand out.
- ✓ Excellent for detecting edema, gliosis, demyelination, and small lesions.
- ✓ Improved lesion conspicuity in MS, stroke, tumor, and infection.
- ✓ Reduces CSF "shine-through" that can hide pathology on T2-weighted images.
- ✓ Same T2 contrast as SE/FSE, just with CSF nulled.

★ **FLAIR = T2 contrast with CSF removed.**

3 WHAT A NORMAL IMAGE GENERALLY LOOKS LIKE (1.5T BRAIN)



KEY CONCEPT

- A 180° inversion pulse flips magnetization.
- After a specific Inversion Time (TI), the longitudinal magnetization of the target tissue passes through zero (null point).
- A 90° pulse then tips the remaining magnetization into the transverse plane for the echo.

- ↑ Increase / Improve
- ↓ Decrease / Reduce
- No / Minimal Effect

4 KEY PARAMETERS AND THEIR EFFECTS

Parameter	Increase	Decrease	Effect on FLAIR
TI (Inversion Time)	Later null point (selected tissue nulled at later TI)	Earlier null point (selected tissue nulled earlier)	Sets which tissue is nulled. Too short → incomplete CSF suppression. Too long → CSF returns and looks bright.
TR	More T1 recovery Higher SNR	Less T1 recovery Lower SNR	Contrast: minor SNR: ↑ with longer TR
TE	More T2 weighting	Less T2 weighting	Affects T2 contrast (and scan time)
ETL (Echo Train Length)	Faster scan Higher SNR More T2 blurring	Slower scan Lower SNR Less T2 blurring	Trade-off between speed and blurring
Slice Thickness	Higher SNR	Lower SNR	Affects SNR

5 OTHER INVERSION RECOVERY APPLICATIONS (CHANGE TI TO NULL DIFFERENT TISSUES)

Sequence	Target Nulled	Typical TI @ 1.5T*	Common Uses
FLAIR	CSF	2200–2500 ms	MS, stroke, tumor, infection, subarachnoid blood
STIR	Fat	140–180 ms	Musculoskeletal edema, bone marrow, fat suppression
Short TI IR	Short T1 tissues (fat, some fluids)	80–120 ms	General IR, subdural hematoma (age dependent)
Double IR	CSF then Fat	TI1 – 2500 ms TI2 – 150 ms	MRCP, black-blood, soft tissue applications

*TI values vary with field strength, vendor and sequence.

REMEMBER

Choose TI to null the unwanted tissue.
Everything else retains T2 contrast.

6 COMMON PROBLEMS AND WHAT THEY INDICATE

Problem / Artifact	What You See	Likely Cause (Engineering View)	What It May Indicate
Incomplete CSF Suppression	CSF appears gray or bright	TI not correct; patient T1 variation; B1 inhomogeneity; incorrect sequence selection	Review TI; check protocol; verify sequence and patient field strength
Over-Suppression	CSF very dark but lesions also dark	TI too long; recovery of CSF magnetization	Shorten TI
Variable Suppression Across Image	One side brighter than other	B1 inhomogeneity; poor shimming; coil sensitivity variation	Check shim; verify coil; check dielectric pads
Residual Ringing (Echo Train)	Ringling lines or blurring around bright CSF spaces	Long ETL, high BW, gradient system issues	Adjust ETL/BW; check gradients

7 ENGINEER CHECKLIST (FLAIR / IR)

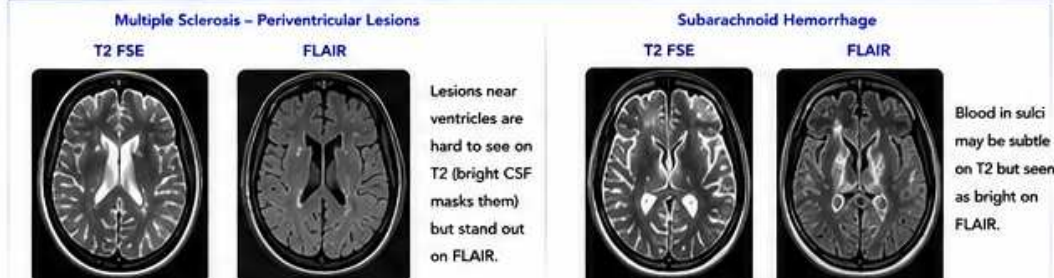
- ✓ Is the correct sequence (FLAIR/IR) selected for the clinical goal?
- ✓ Is TI appropriate for the tissue to be nulled (CSF for FLAIR)?
- ✓ Is CSF properly suppressed (dark in ventricles, sulci)?
- ✓ Is TR, TE and ETL appropriate for the needed contrast and SNR?
- ✓ Is FOV adequate and phase direction optimal (A–P vs P–A)?
- ✓ Is bandwidth appropriate to minimize artifacts?
- ✓ Are gradients and shim performing within specs?
- ✓ Is the coil functioning and positioned correctly?
- ✓ Do artifacts match a known cause (B1, shim, motion, flow, etc.)?
- ✓ Repeat test or sequence to confirm if needed.

QUICK REFERENCE: TI NULL POINTS (1.5T Brain)

Fat (STIR)	140 – 180 ms
Short TI (general)	80 – 120 ms
Gray Matter	~600 – 900 ms
White Matter	~900 – 1200 ms
CSF (FLAIR)	2200 – 2500 ms

Values vary with field strength, vendor and sequence.

8 CLINICAL EXAMPLES (WHY FLAIR WORKS BETTER)



9 PRACTICAL TIPS

- ⚙ Use FLAIR when pathology is near CSF spaces.
- 🔍 Check TI if CSF is not fully suppressed.
- 🔄 Use A → P phase to reduce flow artifacts in many brain exams.
- 📡 Consider parallel imaging to reduce scan time.
- 👁 Always compare with T2 and other sequences.

REMEMBER

FLAIR is T2 contrast with CSF nulled.
The power of inversion recovery is the ability to choose which tissue is dark. Get TI right, and the pathology will often stand out.

KEY TAKEAWAY

Inversion Recovery uses TI to null specific tissue. FLAIR (long TI) suppresses CSF; STIR (short TI) suppresses fat. Understanding TI is key to troubleshooting and optimizing image quality.